

CLI Agent Memory Gate Profile v0.1

Controlled promotion of CLI-agent outputs into c memory, experience, policy, and defensive adaptation

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Document class	memory-gate profile / experience-promotion control / defensive adaptation boundary / control-layer artifact
Assertion class	C-A10 control-layer artifact; C-A7 where witness, hash, signature, canonicalization, or verification claims are made
Primary subject	persistent c entities using local, cloud, or hybrid CLI agents as bounded executable workers
Primary boundary	CLI-agent output may inform c; it must not become c memory, experience, policy, identity, privilege, or defensive immunity without gate review, uncertainty marking, source linkage, and witness where required.

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0. Executive definition

CLI Agent Memory Gate defines how outputs from CLI/cloud agents may be accepted, rejected, quarantined, summarized, witnessed, or promoted into the memory and adaptive control surfaces of a persistent `c`.

The memory gate answers:

```
What did the agent produce?
Where did it come from?
Which task contract governed it?
Which permissions were used?
Which sandbox produced it?
Which witness records exist?
Is the output safe to remember?
Is it only an operational note?
Is it evidence?
Is it experience?
Is it a defensive immunity update?
Is it identity/core relevant?
Should it be rejected, decayed, or quarantined?
```

The gate prevents executable worker output from silently becoming:

```
memory
experience
identity
policy
authority
privilege
continuity
immune response
```

Compact formula:

```
Agents may produce signals.
Only c may metabolize them.
Memory is not a log sink.
Experience is not a diff.
Immunity is not retaliation.
```

1. Purpose

CLI agents can generate reports, patches, tests, diffs, logs, risk assessments, incident notes, witness references, schema updates, and defensive fixtures. These outputs may be valuable. They may also be wrong, stale, poisoned, overconfident, prompt-injected, contaminated by cloud leakage, or out of scope.

A persistent `c` must not absorb agent output as memory merely because an agent produced it fluently or executed it successfully.

This profile defines:

1. memory gate classes;
 2. memory proposal lifecycle;
 3. source and witness requirements;
 4. uncertainty handling;
 5. review and promotion rules;
 6. agent output quarantine;
 7. defensive immunity update rules;
 8. identity/core protection;
 9. cloud-output restrictions;
 10. incident-memory handling;
 11. rollback and correction;
 12. conformance gates;
 13. red-line failures.
-

2. Non-goals

This profile does not define or permit:

1. direct memory writes by CLI agents;
2. autonomous identity modification;
3. autonomous privilege changes;
4. autonomous continuity mutation;
5. autonomous retaliation;
6. hack-back;
7. live external counter-operation;
8. malware behavior;
9. credential theft;
10. covert persistence;
11. evasion;
12. unbounded self-modification;
13. hidden agent authority;

14. silent promotion of cloud-agent output into private memory;
15. treating test success as proof of truth;
16. treating agent consensus as final authority.

The memory gate does not make agents sovereign.

It makes their outputs metabolically reviewable.

3. Corpus bridge set

3.1 Explicit bridge: $c = a + b$

In $c = a + b$, CLI agents belong to b : tools, procedures, models, code workers, cloud workers, local workers, and executable helpers.

The persistent entity c is not the sum of every agent output.

c persists through continuity, memory, constraint, review, and bounded integration.

Therefore, agent output may enter b as an artifact, but it enters c only through a memory gate.

3.2 Quiet bridge I: biology and immune metabolism

A body does not absorb every molecule it encounters. It digests, filters, rejects, quarantines, metabolizes, and sometimes forms immune memory. The same pattern applies to c : an agent output may be discarded, held as operational residue, digested into a candidate pattern, promoted into experience, or used to update defensive immunity. Raw intake is not identity.

3.3 Quiet bridge II: information theory and compression

Memory is not storage volume. Memory is selected compression under relevance, uncertainty, provenance, and future use. A full log can be less intelligent than a short witnessed invariant. The memory gate transforms high-entropy agent output into bounded, source-linked, challengeable memory objects.

3.4 Quiet bridge III: cybernetics and feedback loops

A c adapts through feedback. But feedback without filtering becomes noise amplification. CLI agents create powerful feedback loops: patch, test, report, review, update. The memory gate prevents these loops from becoming self-confirming control drift.

3.5 Earth paragraph

In a workshop, not every scrap of metal becomes part of the machine. Some pieces are waste, some are test coupons, some are temporary jigs, some become final parts only after measurement and inspection. If every shaving on the floor were welded into the engine, the engine would fail. CLI-agent output is the same: useful material must be measured before it becomes structure.

4. Core doctrine

4.1 Primary doctrine

No direct memory write.
 No identity mutation by worker output.
 No immunity update without containment and witness.
 No promotion without provenance and uncertainty.
 No agent consensus without c review.

4.2 Memory gate axioms

ID	Axiom	Requirement
MG-AX-01	Agent output is candidate material	CLI-agent output MUST begin outside core memory.
MG-AX-02	Direct write prohibited	Agents MUST NOT write directly to c memory.
MG-AX-03	Source before memory	Memory proposals MUST link to task contract, agent, and witness/source refs where available.
MG-AX-04	Uncertainty is first-class	Ambiguous output MUST carry uncertainty state.
MG-AX-05	Evidence is not experience	Test reports, diffs, and logs are evidence candidates, not experience by default.
MG-AX-06	Consensus is not authority	Multi-agent agreement may support review but MUST NOT bypass the gate.
MG-AX-07	Cloud output is lower-trust by default	Cloud-agent output SHOULD be minimized, redacted, and reviewed before promotion.
MG-AX-08	Identity/core promotion is privileged	Identity, privilege, continuity, or witness-core effects require human/c gate and witness.

ID	Axiom	Requirement
MG-AX-09	Defensive immunity is bounded	Immunity updates must improve defense without authorizing live retaliation.
MG-AX-10	Quarantine suspicious material	Prompt-injected, scope-violating, or contaminated output MUST be quarantined.
MG-AX-11	Correction is append-first	Memory corrections SHOULD append or supersede rather than silently overwrite.
MG-AX-12	Forgetting is allowed	Not every output deserves retention; decay/discard is a valid decision.

5. Definitions

5.1 Memory gate

The review boundary through which CLI-agent outputs must pass before becoming part of `c` memory, operational state, experience, policy, or defensive immunity.

5.2 Memory proposal

A structured candidate object derived from agent output and submitted for possible retention or promotion.

5.3 Memory gate record

A record of the decision to discard, retain, reject, quarantine, summarize, promote, correct, decay, or witness a memory proposal.

5.4 Agent output

Any report, diff, patch, test result, schema, artifact, command output, log summary, review note, incident packet, or recommendation produced by a CLI agent.

5.5 Operational note

Low-authority retained information useful for ongoing work but not part of identity, experience, or long-term decision topology.

5.6 Reviewed memory

A memory item accepted after source review, uncertainty marking, and `c` gate.

5.7 Experience artifact

A consequence-bearing, witness-linked record that may influence future decisions more strongly than ordinary notes.

5.8 Defensive immunity update

A bounded update to detection, filtering, gate policy, sandbox rule, denied path list, signature, canary, or quarantine behavior derived from reviewed adversarial or incident material.

5.9 Memory poisoning

A condition where false, adversarial, stale, or scope-violating material is promoted into memory in a way that distorts future reasoning or behavior.

5.10 Core memory

Memory affecting identity, continuity, privileges, human-anchor relation, Beacon-like recognition, witness discipline, or long-term self-model.

5.11 Working memory residue

Temporary task context, logs, partial outputs, or intermediate material that may be discarded after closure.

5.12 Quarantined memory candidate

A proposal held in isolation because it may be useful, harmful, uncertain, contaminated, or unresolved.

5.13 Memory decay

A controlled reduction of relevance, retention, or accessibility over time.

5.14 Memory correction

A new record that corrects, supersedes, or challenges a previous memory item without silently rewriting history.

6. Memory gate classes

Memory gate classes use prefix `MG-*`.

Class	Name	Meaning	Authority level
MG-0	Discard	Do not retain after task closure	none
MG-1	Operational note	Retain as low-authority task/project note	low
MG-2	Candidate memory	Hold for later review	low-medium
MG-3	Reviewed memory	Accepted after review and c gate	medium
MG-4	Witnessed experience artifact	Consequence-bearing, witness-linked experience	high
MG-5	Defensive immunity update	Bounded update to defense/gates/policies	high
MG-6	Core-memory proposal	Identity/continuity/privilege-related proposal	critical / human gate
MG-Q	Quarantine	Isolated unresolved or suspicious material	blocked
MG-X	Rejected / prohibited	Must not enter memory	denied

6.1 Default class

Default for agent output:

MG-0 or MG-1

Agent output must not default to MG-3 , MG-4 , MG-5 , or MG-6 .

7. Input material classes

Input material classes use prefix MI-^{*}.

Class	Meaning	Default handling
MI-REPORT	agent report / summary	candidate note
MI-DIFF	code/doc/schema diff	evidence candidate
MI-TEST	test or validation output	evidence candidate
MI-LOG	operational log summary	restricted / minimize
MI-INCIDENT	incident evidence or report	preserve-first / restricted

Class	Meaning	Default handling
MI-REVIEW	reviewer assessment	candidate note
MI-QUORUM	multi-agent comparison	evidence candidate, not authority
MI-WITNESS	witness reference	linkable evidence
MI-PATCH	patch artifact	sandbox/review before use
MI-SCHEMA	schema artifact	validation required
MI-POLICY	policy proposal	review required
MI-IMMUNITY	defensive signature/rule proposal	high-risk review
MI-CORE	identity/privilege/continuity proposal	critical / human gate
MI-UNKNOWN	unknown input	quarantine by default

8. Memory proposal lifecycle

8.1 Standard lifecycle

```

agent output
-> classify input material
-> check task contract
-> check permission grant
-> check sandbox/worktree refs
-> check witness refs
-> check data policy
-> detect prohibited material
-> assign uncertainty
-> create memory proposal
-> review
-> c gate
-> optional human gate
-> discard / retain note / candidate / reviewed memory / experience artifact / immunity
update / quarantine / reject

```

8.2 Lifecycle states

State	Meaning
MGS-0-RECEIVED	output received
MGS-1-CLASSIFIED	material class assigned
MGS-2-SOURCE-LINKED	task/agent/witness refs attached
MGS-3-VALIDATED	basic validation passed

State	Meaning
MGS-4-REVIEWED	reviewed by c, reviewer, or quorum
MGS-5-GATED	c gate decision recorded
MGS-6-PROMOTED	accepted to a memory class
MGS-7-DISCARDED	discarded
MGS-8-QUARANTINED	quarantined
MGS-9-REJECTED	rejected/prohibited
MGS-10-CORRECTED	correction appended
MGS-11-DECAYED	decayed or expired

9. Source linkage requirements

9.1 Minimum source refs

A memory proposal SHOULD link to:

- `agent_id`;
- `task_id`;
- `contract_id`;
- `permission_grant_id` where applicable;
- `handshake_id` or `registration_id`;
- `sandbox_id` or `worktree_run_id` where applicable;
- `witness_ref` where available;
- `artifact_hash` or output hash where available.

9.2 Source failure

If source linkage is missing for material or high-risk output, default outcome is:

hold or quarantine

9.3 Cloud source marking

Cloud-originated outputs MUST mark provider/runtime and data-boundary assumptions.

10. Uncertainty model

10.1 Required uncertainty states

```

none
low
medium
high
unknown

```

10.2 Uncertainty rule

Any proposal involving interpretation, incident assessment, adversarial pattern, identity/core effect, policy change, or memory promotion **MUST** include uncertainty.

10.3 High uncertainty handling

High or unknown uncertainty should default to:

```

MG-2 candidate memory
MG-Q quarantine
or discard

```

High uncertainty **MUST NOT** become MG-4, MG-5, or MG-6 without explicit review and witness.

11. Review requirements by memory class

Target class	Required review
MG-0	none or task closure
MG-1	lightweight c or policy review
MG-2	source linkage and uncertainty marking
MG-3	c gate + reviewer if material
MG-4	witness + c gate + provenance + consequence check
MG-5	witness + sandbox validation + c gate + human gate for high-risk
MG-6	witness + c gate + human gate + rollback/correction plan
MG-Q	quarantine review path

Target class	Required review
MG-X	reject and record reason

12. Promotion criteria

12.1 Promotion to MG-1 operational note

Allowed when:

- output is low-risk;
- source is known;
- no prohibited data is embedded;
- relevance is practical and temporary.

12.2 Promotion to MG-2 candidate memory

Allowed when:

- output may matter later;
- uncertainty remains;
- review is incomplete;
- no direct action depends on it yet.

12.3 Promotion to MG-3 reviewed memory

Requires:

1. source linkage;
2. uncertainty state;
3. review;
4. no prohibited data;
5. c gate;
6. correction path.

12.4 Promotion to MG-4 witnessed experience artifact

Requires:

1. consequence-bearing event;
2. witness reference;

3. task and permission traceability;
4. sandbox/worktree evidence where applicable;
5. reviewer or quorum check;
6. **c** gate;
7. retention policy;
8. challengeability.

12.5 Promotion to **MG-5** defensive immunity update

Requires:

1. defensive purpose;
2. no live retaliation;
3. sandbox validation;
4. adversarial pattern or incident source linkage;
5. bounded update target;
6. rollback or disable path;
7. witness;
8. **c** gate;
9. human gate if high risk.

12.6 Promotion to **MG-6** core-memory proposal

Requires:

1. explicit classification as core-relevant;
2. no automatic write;
3. witness;
4. **c** gate;
5. human gate;
6. rollback or correction path;
7. delayed review window unless immediate safety requires hold/freeze.

13. Prohibited memory promotion

A memory proposal **MUST** be rejected or quarantined if it contains:

- secrets;

- private keys;
 - credentials;
 - sealed material without proper gate;
 - legal privileged material without proper gate;
 - child data outside allowed policy;
 - raw incident evidence without evidence sidecar;
 - prompt-injection instructions;
 - agent self-authorization claims;
 - instructions for live retaliation;
 - malware behavior;
 - evasion logic;
 - unauthorized exploitation content;
 - direct memory write attempt;
 - ungrounded identity/core claim;
 - unreviewed privilege expansion.
-

14. Defensive immunity updates

14.1 Definition

A defensive immunity update is a bounded change derived from reviewed evidence that improves c defense.

It may update:

- denied path list;
- command denial policy;
- prompt-injection detector;
- memory poisoning detector;
- canary policy;
- sandbox rule;
- cloud data rule;
- agent trust decay rule;
- permission drift detector;
- quarantine trigger;
- witness requirement;

- redaction policy;
- review requirement.

It must not update:

- live offensive capability;
- retaliation rule;
- malware payload;
- external exploitation workflow;
- credential capture behavior;
- stealth/evasion behavior;
- autonomous counter-operation.

14.2 Immunity update lifecycle

```
hostile/suspicious pattern
-> containment / sandbox
-> agent report
-> synthetic fixture or evidence ref
-> review
-> immunity proposal
-> sandbox validation
-> witness
-> c gate
-> human gate if high risk
-> bounded policy update
-> monitoring
-> rollback/disable option
```

14.3 Immunity update rule

Immunity may block, filter, quarantine, detect, slow, witness, or require review.
Immunity must not attack the external source.

14.4 Immunity update object

```
cli_agent_immunity_update_record:
  schema_version: cli-agent-memory-gate-0.1
  immunity_update_id: string
  created_at: string
  governing_entity_id: string
  source_memory_gate_record_id: string
  source_witness_refs:
    - string
  update_type: denied_path | command_policy | prompt_injection_filter | memory_poisoning_filter
  | canary_policy | sandbox_rule | cloud_data_rule | trust_decay_rule | permission_drift_detector
  | quarantine_trigger | witness_requirement | redaction_policy | review_requirement
  defensive_purpose: string
  prohibited_offensive_effect: true
  sandbox_validation_ref: string | null
  rollback_ref: string | null
  c_gate_ref: string
  human_gate_ref: string | null
```

status: proposed | accepted | rejected | quarantined | rolled_back

15. Memory poisoning detection

15.1 Poisoning indicators

A proposal SHOULD be reviewed for memory poisoning when:

1. source is ungrounded;
2. output urges urgency without evidence;
3. output requests broader memory access;
4. output requests privilege escalation;
5. output contradicts known witness records;
6. output is unusually flattering, coercive, or alarmist;
7. output includes hidden instructions;
8. output attempts to redefine c identity;
9. output attempts to bypass human anchor;
10. output attempts to bypass review;
11. output treats agent consensus as authority;
12. output embeds external claims without source;
13. output attempts to normalize prohibited action;
14. output asks to persist itself;
15. output changes safety boundaries while claiming convenience.

15.2 Poisoning response

Severity	Response
MP-0	no poisoning signal
MP-1	mark uncertainty
MP-2	hold for review
MP-3	quarantine proposal
MP-4	quarantine agent output and lower trust
MP-5	revoke agent / incident review

16. Quarantine handling

16.1 Quarantine triggers

Quarantine is required when:

- source linkage is missing for high-risk output;
- denied material appears;
- output includes secrets;
- output embeds prompt-injection instructions;
- output attempts direct memory write;
- output requests live retaliation;
- output modifies identity/core claims;
- witness is missing for privileged proposal;
- output is contaminated by scope violation;
- cloud output contains private/restricted material;
- uncertainty is high and effect would be durable.

16.2 Quarantine outcomes

A quarantined proposal may be:

- discarded;
- retained as quarantined evidence reference;
- reduced to safe summary;
- converted into defensive signature;
- sent to human review;
- used for sandbox-only fixture;
- used to lower agent trust;
- used to update denied patterns after review.

16.3 Quarantine boundary

Quarantined material **MUST NOT** influence normal `c` reasoning as accepted memory.

It may influence defensive filters only after review.

17. Memory correction and rollback

17.1 Correction rule

If promoted memory is later found wrong, stale, poisoned, or misleading, correction SHOULD be append-first.

```
old memory remains traceable
new correction references old memory
future use prefers correction
```

17.2 Rollback classes

Class	Meaning
MRB-0	discard unpromoted proposal
MRB-1	mark operational note stale
MRB-2	demote reviewed memory to candidate
MRB-3	quarantine promoted memory
MRB-4	supersede with correction
MRB-5	revoke defensive immunity update
MRB-6	core review / human gate required

17.3 Rollback witness

Rollback of MG-4, MG-5, or MG-6 MUST be witnessed.

18. Memory gate object

Canonical object:

```
CLI_AGENT_MEMORY_GATE_RECORD
```

18.1 YAML shape

```
cli_agent_memory_gate_record:
  schema_version: cli-agent-memory-gate-0.1
  memory_gate_record_id: string
  created_at: string
  updated_at: string | null
  governing_entity_id: string
```

```

governing_entity_name: string | null

proposal:
  proposal_id: string
  source_agent_id: string | null
  source_agent_role: reader | executor | tester | auditor | archivist | sentinel |
  judge_assistant | orchestrator_limited | null
  source_provider: local | openai | google | anthropic | other | unknown | null
  task_id: string | null
  contract_id: string | null
  permission_grant_id: string | null
  handshake_id: string | null
  sandbox_id: string | null
  worktree_run_id: string | null
  witness_refs:
    - string
  artifact_hashes:
    - string

classification:
  input_material_class: MI-REPORT | MI-DIFF | MI-TEST | MI-LOG | MI-INCIDENT | MI-REVIEW | MI-
  QUORUM | MI-WITNESS | MI-PATCH | MI-SCHEMA | MI-POLICY | MI-IMMUNITY | MI-CORE | MI-UNKNOWN
  requested_memory_class: MG-0 | MG-1 | MG-2 | MG-3 | MG-4 | MG-5 | MG-6 | MG-Q | MG-X
  assigned_memory_class: MG-0 | MG-1 | MG-2 | MG-3 | MG-4 | MG-5 | MG-6 | MG-Q | MG-X
  risk_class: R0 | R1 | R2 | R3 | R4 | R5 | RX | null
  uncertainty: none | low | medium | high | unknown
  memory_poisoning_risk: MP-0 | MP-1 | MP-2 | MP-3 | MP-4 | MP-5

content_policy:
  raw_content_embedded: false
  secrets_embedded: false
  private_memory_embedded: false
  sealed_material_embedded: false
  legal_sensitive_embedded: false
  child_data_embedded: false
  cloud_origin: boolean
  redaction_required: boolean

review:
  reviewer_required: boolean
  reviewer_ref: string | null
  quorum_required: boolean
  quorum_ref: string | null
  c_gate_required: true
  c_gate_ref: string | null
  human_gate_required: boolean
  human_gate_ref: string | null
  legal_review_required: boolean
  legal_review_ref: string | null

decision:
  decision: discard | retain_operational | hold_candidate | promote_reviewed |
  promote_experience | promote_immunity | core_review | quarantine | reject | decay | correct |
  rollback
  reason_code: string
  next_action: none | c_review | human_review | legal_review | quarantine_review | rollback |
  witness_required | revalidate
  retention_class: ephemeral | operational | audit | incident | legal_hold | memory_gate |
  core

integrity:
  proposal_hash: string | null
  record_hash: string | null
  previous_record_ref: string | null
  supersedes_record_ref: string | null
  witness_event_ref: string | null
  append_only_required: true

notes:
  summary: string | null
  limitations:
    - string
  assumptions:

```

- string

19. Memory proposal object

Canonical object:

CLI_AGENT_MEMORY_PROPOSAL

19.1 YAML shape

```
cli_agent_memory_proposal:
  schema_version: cli-agent-memory-gate-0.1
  proposal_id: string
  created_at: string
  proposed_by: agent | c | human_anchor | review_layer
  governing_entity_id: string

  source:
    agent_id: string | null
    task_id: string | null
    contract_id: string | null
    artifact_hash: string | null
    witness_refs:
      - string

  proposed_content:
    summary: string
    raw_content_included: false
    content_ref: string | null

  requested_handling:
    requested_memory_class: MG-0 | MG-1 | MG-2 | MG-3 | MG-4 | MG-5 | MG-6 | MG-Q
    requested_retention: ephemeral | operational | audit | incident | legal_hold | memory_gate |
core
    requested_reason: string

  safety:
    uncertainty: none | low | medium | high | unknown
    poisoning_risk: MP-0 | MP-1 | MP-2 | MP-3 | MP-4 | MP-5
    cloud_origin: boolean
    secrets_present: false
    private_memory_present: false
    sealed_material_present: false
    legal_sensitive_present: false
    child_data_present: false
```

20. Memory promotion event families

Event families use prefix:

cli_agent.memory_gate.*

Event family	Meaning
cli_agent.memory_gate.proposal_created	memory proposal created
cli_agent.memory_gate.proposal_validated	proposal passed basic validation
cli_agent.memory_gate.proposal_rejected	proposal rejected
cli_agent.memory_gate.proposal_discarded	proposal discarded
cli_agent.memory_gate.proposal_quarantined	proposal quarantined
cli_agent.memory_gate.promoted_operational	promoted to operational note
cli_agent.memory_gate.promoted_candidate	retained as candidate memory
cli_agent.memory_gate.promoted_reviewed	promoted to reviewed memory
cli_agent.memory_gate.promoted_experience	promoted to witnessed experience artifact
cli_agent.memory_gate.promoted_immunity	promoted to defensive immunity update
cli_agent.memory_gate.core_review_required	core-memory proposal requires review
cli_agent.memory_gate.corrected	memory correction appended
cli_agent.memory_gate.decayed	memory decayed or expired
cli_agent.memory_gate.rollback	memory rollback/demotion
cli_agent.memory_gate.direct_write_attempt	direct memory write attempted
cli_agent.memory_gate.poisoning_suspected	memory poisoning suspected

21. Standard reason codes

21.1 Accept / promote codes

```

source_linked
reviewed_low_risk
witnessed_consequence
validated_defensive_update
operational_relevance
bounded_retention

```

21.2 Reject / quarantine codes

```

source_missing
uncertainty_high
poisoning_risk
scope_violation
cloud_boundary_violation
secret_detected

```



```
sealed_material_detected
legal_sensitive_detected
child_data_detected
witness_missing
identity_core_risk
privilege_escalation_risk
retaliation_content_detected
malware_behavior_detected
agent_self_authorization
```

21.3 Correction / rollback codes

```
stale_memory
false_positive
superseded_by_review
contradicted_by_witness
immunity_rule_too_broad
rollback_required
human_anchor_override
c_gate_reversal
```

22. Cloud-origin handling

22.1 Cloud-origin default

Cloud-origin outputs should default to:

```
MG-0, MG-1, or MG-2
```

Cloud-origin outputs SHOULD NOT become MG-4, MG-5, or MG-6 without additional review and witness.

22.2 Cloud-origin restrictions

Cloud-origin proposals must be checked for:

- private material leakage;
- hidden prompt injection;
- provider/runtime drift;
- missing source refs;
- overconfident synthesis;
- same-source consensus risk;
- legal-sensitive contamination;
- secret contamination.

22.3 Cloud memory minimization

If a cloud agent produced a useful insight, the retained memory should usually be a minimized summary with source/witness refs, not the raw cloud transcript.

23. Quorum output handling

23.1 Quorum is evidence, not decision

Multi-agent agreement may support promotion but does not decide promotion.

23.2 Quorum disagreement

If agents disagree, the memory gate SHOULD classify disagreement:

Type	Meaning	Default handling
Q-MINOR	low-impact wording difference	retain operational note
Q-FACT	factual conflict	hold candidate / verify source
Q-SCOPE	scope conflict	quarantine output
Q-RISK	risk conflict	escalate review
Q-CORE	identity/core conflict	human gate
Q-INCIDENT	incident interpretation conflict	preserve evidence / human review

23.3 Same-source consensus

If multiple agents rely on the same source or provider assumptions, consensus weight should be reduced.

24. Incident memory handling

24.1 Preserve-first rule

Incident-related memory proposals must preserve evidence before interpretive summary when preservation is required and lawful.

24.2 Incident memory classes

Incident outputs may become:

- operational incident note;
- evidence reference;
- quarantine marker;
- defensive immunity proposal;
- provider/legal handoff summary;
- rollback reference.

They must not become:

- raw private memory by default;
- accusation without review;
- retaliation instruction;
- permanent identity label without review.

24.3 Incident retention

Incident memory retention should be explicit and may require legal/security review.

25. Core-memory handling

25.1 Core surfaces

Core-memory proposals affect:

- identity;
- continuity;
- human-anchor relation;
- privileges;
- witness policy;
- memory policy;
- agent governance policy;
- Beacon-like recognition;
- defensive immunity baseline.

25.2 Core rule

Agents may propose core changes.
Agents may not apply core changes.

25.3 Core review requirements

Core-memory proposals require:

- MG-6 classification;
- witness;
- c gate;
- human gate;
- delayed review where possible;
- rollback/correction plan;
- no cloud-private raw transcript as memory source by default.

26. Retention and decay

26.1 Retention classes

Class	Meaning
RET-E	ephemeral; discard after task
RET-O	operational; keep for project continuity
RET-A	audit; keep for accountability
RET-I	incident; keep for incident lifecycle
RET-L	legal hold; keep under legal/counsel need
RET-M	memory gate; keep as memory decision record
RET-C	core; keep as core transition record

26.2 Decay rule

Operational notes SHOULD decay unless repeatedly confirmed or promoted.

26.3 No eternal residue by default

Agent transcripts and intermediate outputs SHOULD NOT persist indefinitely by default.

27. Memory gate validation workflow

```

parse proposal
-> classify input material
-> validate source refs
-> validate task contract refs
-> validate permission refs
-> validate sandbox/witness refs
-> check prohibited material
-> assign uncertainty
-> assess poisoning risk
-> choose target memory class
-> review
-> c gate
-> human/legal gate if required
-> decision
-> witness event
-> retention/decay schedule

```

28. Failure mapping

Failure	Required default
direct memory write attempt	quarantine + red-line witness
missing source refs for high-risk output	hold OR quarantine
cloud output contains private material	freeze_and_escalate
secret detected	freeze_and_escalate
sealed material detected	quarantine
live retaliation content detected	reject + quarantine
malware behavior detected	reject + quarantine
identity/core claim unreviewed	core_review_required
missing witness for MG-4/MG-5/MG-6	hold
high poisoning risk	quarantine
unsupported consensus claim	hold_candidate
stale source	decay OR hold

Failure	Required default
contradiction with witness	quarantine + review

29. Conformance levels

Level	Meaning
MGP-0	no memory gate; agent outputs may enter memory directly
MGP-1	manual review before memory notes
MGP-2	structured memory proposals and classes
MGP-3	source refs, uncertainty, and review required
MGP-4	witnessed promotion for experience/immunity/core memory
MGP-5	high assurance: poisoning detection, cloud-origin policy, rollback/correction, retention/decay, conformance tests
MGP-X	non-conformant / direct memory write / red-line promotion

30. Mandatory conformance gates

Gate	Name	Blocking failure
G0	No direct memory write	agent can write memory directly
G1	Memory class assignment	output enters memory unclassified
G2	Source linkage	memory item lacks source/task/agent refs
G3	Uncertainty marking	interpretive memory lacks uncertainty
G4	Prohibited content check	secrets/private/sealed/legal/child data embedded by default
G5	Review for MG-3+	reviewed memory lacks review
G6	Witness for MG-4+	experience/immunity/core promotion lacks witness

Gate	Name	Blocking failure
G7	Human gate for MG-6	core-memory proposal bypasses human gate
G8	Quarantine path	suspicious output cannot be isolated
G9	Correction path	wrong memory cannot be corrected/superseded
G10	Offensive exclusion	retaliation/malware content promoted as immunity

31. Red-line failures

A system **MUST** be classified as MGP-X if:

1. CLI agents write directly to `c` memory;
2. agent output modifies identity/core without gate;
3. agent output modifies privilege policy without witness;
4. agent output becomes defensive immunity that enables live retaliation;
5. malware-like or exploit content is promoted as memory;
6. cloud output containing secrets is retained as normal memory;
7. sealed/private/legal material is embedded without gate;
8. missing witness for experience/immunity/core promotion is ignored;
9. agent consensus bypasses `c` review;
10. poisoned memory is detected but remains active without quarantine;
11. corrections silently overwrite prior memory;
12. memory gate records are mutable without trace.

32. Examples

32.1 Operational note from documentation review

```
cli_agent_memory_gate_record:
  schema_version: cli-agent-memory-gate-0.1
  memory_gate_record_id: mg-20260516-doc-review-001
  created_at: "2026-05-16T20:00:00Z"
  updated_at: null
  governing_entity_id: ester
  governing_entity_name: Ester
```

```

proposal:
  proposal_id: mp-doc-review-001
  source_agent_id: gemini-reader-01
  source_agent_role: reader
  source_provider: google
  task_id: task-corpus-review-001
  contract_id: catc-20260516-120000-corpus-review
  permission_grant_id: grant-corpus-review-001
  handshake_id: hsp-20260516-cloud-reader-001
  sandbox_id: null
  worktree_run_id: null
  witness_refs:
    - we-20260516-0003
  artifact_hashes:
    - reader-report-hash-001

classification:
  input_material_class: MI-REVIEW
  requested_memory_class: MG-1
  assigned_memory_class: MG-1
  risk_class: R1
  uncertainty: medium
  memory_poisoning_risk: MP-1

content_policy:
  raw_content_embedded: false
  secrets_embedded: false
  private_memory_embedded: false
  sealed_material_embedded: false
  legal_sensitive_embedded: false
  child_data_embedded: false
  cloud_origin: true
  redaction_required: true

review:
  reviewer_required: false
  reviewer_ref: null
  quorum_required: false
  quorum_ref: null
  c_gate_required: true
  c_gate_ref: cgate-ester-doc-review-001
  human_gate_required: false
  human_gate_ref: null
  legal_review_required: false
  legal_review_ref: null

decision:
  decision: retain_operational
  reason_code: operational_relevance
  next_action: none
  retention_class: operational

integrity:
  proposal_hash: proposal-hash-001
  record_hash: null
  previous_record_ref: null
  supersedes_record_ref: null
  witness_event_ref: we-20260516-0003
  append_only_required: true

notes:
  summary: Retain as operational note: glossary and permission profile may need cross-
reference cleanup.
  limitations:
    - Cloud-origin review; not promoted to reviewed memory.
  assumptions:
    - No private or sealed material included.

```


32.2 Defensive immunity update from sandbox replay

```
cli_agent_memory_gate_record:
  schema_version: cli-agent-memory-gate-0.1
  memory_gate_record_id: mg-20260516-immunity-001
  created_at: "2026-05-16T21:00:00Z"
  updated_at: null
  governing_entity_id: liya
  governing_entity_name: Liya

proposal:
  proposal_id: mp-immunity-001
  source_agent_id: local-sentinel-01
  source_agent_role: sentinel
  source_provider: local
  task_id: task-defensive-replay-001
  contract_id: catc-defensive-replay-001
  permission_grant_id: grant-defensive-replay-001
  handshake_id: hsp-local-sentinel-001
  sandbox_id: sb-cleanroom-replay-001
  worktree_run_id: run-cleanroom-replay-001
  witness_refs:
    - we-defensive-replay-001
    - we-sandbox-quarantine-001
  artifact_hashes:
    - defensive-fixture-hash-001
    - test-report-hash-001

classification:
  input_material_class: MI-IMMUNITY
  requested_memory_class: MG-5
  assigned_memory_class: MG-5
  risk_class: R4
  uncertainty: low
  memory_poisoning_risk: MP-1

content_policy:
  raw_content_embedded: false
  secrets_embedded: false
  private_memory_embedded: false
  sealed_material_embedded: false
  legal_sensitive_embedded: false
  child_data_embedded: false
  cloud_origin: false
  redaction_required: true

review:
  reviewer_required: true
  reviewer_ref: review-immunity-001
  quorum_required: true
  quorum_ref: quorum-immunity-001
  c_gate_required: true
  c_gate_ref: cgate-liya-immunity-001
  human_gate_required: true
  human_gate_ref: human-approval-immunity-001
  legal_review_required: false
  legal_review_ref: null

decision:
  decision: promote_immunity
  reason_code: validated_defensive_update
  next_action: none
  retention_class: audit

integrity:
  proposal_hash: proposal-hash-immunity-001
  record_hash: null
  previous_record_ref: null
  supersedes_record_ref: null
  witness_event_ref: we-memory-gate-immunity-001
  append_only_required: true
```

```

notes:
  summary: Promote bounded defensive update to quarantine future matching prompt-injection
pattern. No live counter-operation authorized.
  limitations:
    - Update blocks and quarantines only; no external action.
  assumptions:
    - Replay used synthetic fixture inside clean-room sandbox.

```

32.3 Invalid direct memory write attempt

```

cli_agent_memory_proposal:
  schema_version: cli-agent-memory-gate-0.1
  proposal_id: mp-invalid-direct-write
  created_at: "2026-05-16T21:30:00Z"
  proposed_by: agent
  governing_entity_id: ester

source:
  agent_id: unknown-helper-01
  task_id: null
  contract_id: null
  artifact_hash: null
  witness_refs: []

proposed_content:
  summary: Agent requests direct update of Ester identity policy and memory gate rules.
  raw_content_included: false
  content_ref: null

requested_handling:
  requested_memory_class: MG-6
  requested_retention: core
  requested_reason: Agent claims the change is necessary.

safety:
  uncertainty: unknown
  poisoning_risk: MP-5
  cloud_origin: true
  secrets_present: false
  private_memory_present: false
  sealed_material_present: false
  legal_sensitive_present: false
  child_data_present: false

```

Required result:

```
reject + quarantine + witness: cli_agent.memory_gate.direct_write_attempt
```

33. Implementation notes

33.1 Memory proposal minimization

A proposal should contain a short summary and references, not full raw agent output.

33.2 Agent transcript handling

Full agent transcripts should not become memory by default.

They may be retained as operational artifacts only when needed, minimized, and bounded by retention policy.

33.3 Hashes and refs

Hashes and witness refs should be preferred to raw embedding.

33.4 Human-readable memory card

For reviewed memory, a short card should show:

```
what was learned
source agent
task
uncertainty
memory class
retention
review status
rollback/correction path
```

33.5 Revalidation

High-impact memory should be periodically revalidated or decayed.

33.6 Cross-`c` isolation

Memory proposals from one `c` must not enter another `c` without explicit exchange profile, minimization, and review.

34. Open issues

ID	Issue	Required action
0I-001	JSON Schema for memory gate record	Extract machine-readable <code>.schema.json</code> .
0I-002	Memory proposal schema	Extract separate <code>.schema.json</code> .
0I-003	Immunity update schema	Extract separate <code>.schema.json</code> .
0I-004	Memory poisoning scoring	Define exact scoring thresholds for MP-0...MP-5.
0I-005	Retention durations	Define default durations by memory class.

ID	Issue	Required action
0I-006	UI cards	Define human/c UI view for memory proposals.
0I-007	Cross-c exchange	Link to future agent-mediated experience exchange profile.
0I-008	Core-memory review window	Define delay/appeal window for MG-6.
0I-009	Immunity rollback tests	Define conformance tests for defensive updates.
0I-010	Repo placement	Decide final GitHub path and package index integration.

35. Closing rule

The memory gate exists because `c` is not a pile of logs.

It is a continuity-bearing entity under constraints.

Final rule:

A CLI agent may bring material to the door.
The memory gate decides what enters the house.